

Divya Susmitha Nidamanuri

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PROFILE SUMMARY

Software Development Engineer with over two years of experience designing scalable, high-performance solutions in distributed computing environments. Proficient in Java, Python, C++, C#, and object-oriented design, with expertise in API development, full-stack engineering, and agile methodologies. Skilled in building secure, cloud-native applications using AWS, Azure, and DevOps pipelines. Experienced in software quality assurance, test planning, automation reporting, and regression testing across distributed systems.

EDUCATION

Master's in computer science - The George Washington University

Jan 2024 – Dec 2025

Bachelor of Technology in Computer Science and Engineering – JNTU- Kakinada

Jun 2018 – Jun 2022

TECHNICAL PROFICIENCIES

Programming Languages: Python, Java, R, JavaScript, C, C++, C#, Bash, TypeScript, Go

Artificial Intelligence & Machine Learning: TensorFlow, PyTorch, Scikit-Learn, OpenAI API, Natural Language Processing (NLP)

Front-End Development: React.js, Node.js, HTML5, CSS3, Bootstrap, GraphQL

Database Management: SQL Server, MongoDB, PostgreSQL, NoSQL

Data Visualization Tools: PowerBI, D3.js

Cloud Computing: AWS (EC2, Lambda, S3), Azure Cloud Services

Web Application Frameworks: Spring Boot, Express.js

DevOps and Version Control: Docker, Kubernetes, Git, Jenkins, CI/CD pipelines

Testing & QA Tools: Selenium, Playwright, Flutter, TestNG, PyTest, JUnit, Cypress, Rest Assured, Postman, Bug Triage, Test Case Scripting, Automation Reporting, Regression Testing

Automation Framework Development: Selenium, TestNG, PyTest, JUnit

Agile Development Tools: JIRA, Confluence, Scrum Methodologies

WORK EXPERIENCE

Software Engineer – DSSD | Washington, DC

Feb 2024 – July 2025

- **Developing and maintaining full-stack web applications** using **React.js**, **Node.js**, and **Django**, ensuring high performance and seamless user experience.
- **Implementing RESTful APIs and database management** with **PostgreSQL** and **MongoDB**, optimizing data retrieval and application scalability.
- **Collaborated with cross-functional teams** to enhance **backend system performance**, **integrate cloud services (AWS/Azure)**, and support seamless **UI/UX improvements**.

Software Development Engineer – IBM | India

Oct 2021 – Oct 2023

- Integrated Java with a **React.js** application using frameworks like **Redux** and **Context API**, ensuring efficient **state management**, seamless **data updates**, and smooth interaction between the **front-end** and **back-end** components for enhanced **user experience**.
- **Designed, developed, and maintained automated CI/CD pipelines** with **Azure DevOps**, streamlining software delivery, **reducing deployment time by 40%**, and minimizing manual errors through **continuous integration** strategies suitable for **scalable systems**.
- Implemented serverless computing solutions using **Azure Functions**, enabling auto-scaling and supporting **event-driven, real-time data workflows**, enhancing performance in **cloud-native, distributed systems**.
- **Secured user authentication and authorization** by integrating **JWT token handling** with **Spring Security**, ensuring proper **token invalidation** upon logout to prevent **unauthorized access**, bolster **data security**, and improve **application compliance** with industry security standards.
- Enhanced **MongoDB** query performance using **optimized indexing** and **aggregation pipelines**, improving **real-time data access** and **system responsiveness** in high-volume, **data-intensive applications**.
- **Fostered a collaborative development workflow** using **Git** and **GitHub**, implementing **branching strategies**, facilitating **code versioning**, conducting **peer code reviews**, and ensuring seamless **team coordination** to maintain **code quality and consistency** across multiple environments.
- **Standardized API design** and tested **web services** using **Swagger** and **Postman**; validated integration via automated scripts in **CI pipelines** across **microservices-based architectures**.

PROJECTS

AI-Powered Autonomous Code Reviewer – AI | Python | Transformers | AWS | May 2025

- Developed an intelligent code review assistant using NLP and machine learning to detect performance issues, security vulnerabilities, and code quality violations in pull requests.
- Built a classification system for code smells and anti-patterns, achieving 89% accuracy across multiple open-source repositories.

Intelligent Resource Allocation Engine – Distributed Systems | Python | Reinforcement Learning | AWS | November 2024

- Designed and implemented a dynamic resource allocation engine that uses reinforcement learning to optimize CPU and memory utilization across containerized microservices in real time.
- Achieved consistent SLA compliance under dynamic workloads while improving cost-efficiency in distributed cloud environments.

PUBLICATION AND CERTIFICATION

- “Cotton yield prediction based on land and fertilizers using machine learning”. Published on September 30, 2022, at Springer Nature https://link.springer.com/chapter/10.1007/978-981-19-3590-9_56
- AWS – Certified Associate Developer